FACULTY OF COMPUTING

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SECP3213 - BUSINESS INTELLIGENCE

SECTION 01

RETAIL SALES AND INVENTORY MANAGEMENT ANALYSIS

PHASE 2: DATA WORKFLOW (ALTERYX DESIGNER)

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1.0 Data Workflow

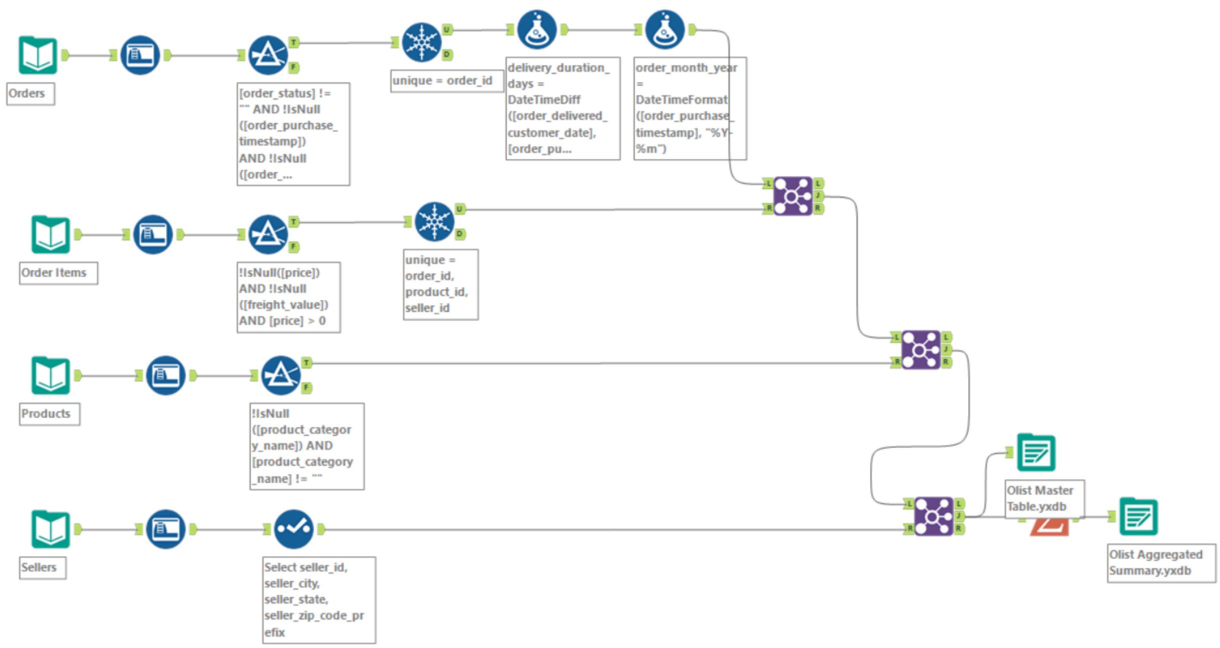

Figure 1: Data Workflow to Create Unified Brazilian E-Commerce Analytical Dataset

Figure 1 shows the data workflow which was developed in Alteryx Designer to perform the extraction, cleaning, integration, and transformation process for the datasets obtained from the Brazilian E-Commerce Public Dataset by Olist. The workflow integrates four separate datasets into a unified master analytical table, which is exported for dashboard development in Microsoft Power BI. The objective of this data workflow is to create an output dataset, ready to be analysed and perform data visualization on sales performance, inventory trends, and supplier operations. The configuration of each node will be explained in detail in the upcoming subtopics.

1.1 Data Input

The input stage begins by importing multiple CSV datasets downloaded from Kaggle into Alteryx Designer using the Input Data Tool. The datasets are loaded directly from local storage directories into the workflow for processing. During import, the input configuration preserves the original dataset formatting and character encoding to ensure Portuguese regional characters and categorical values remain consistent throughout the analytical process. Table 1 shows the dataset overview.

Dataset	Rows	Fields
olist_orders_dataset	99,441	order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date
olist_order_items_dataset	112,650	order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value
olist_products_dataset	32,951	product_id, product_category_name, product_name_lenght, product_description_lenght, product_photos_qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm
olist_sellers_dataset	3,095	seller_id, seller_zip_code_prefix, seller_city, seller_state

Table 1: Overview of Brazilian E-Commerce Order Item Dataset

1.2 Data Cleaning & Data Preparation

Auto Field tool is utilized for each datasets to prevent the attributes to load under generic, memory-heavy V_String types where these nodes automatically optimize allocations down to minimal database structures. Besides, the variable memory allocations for each datasets are compressed into tight, optimized data formats. For instance, the Brazilian E-Commerce Order Item Dataset (olist_order_items_dataset.csv) utilised the Auto Field tool by converting shipping_limit_date field to DateTime and financial fields like price and freight_value to Double as shown in *Figure 2*.

Before

Record	Name	Type	Size
1	order_id	V_String	254
2	order_item_id	V_String	254
3	product_id	V_String	254
4	seller_id	V_String	254
5	shipping_limit_date	V_String	254
6	price	V_String	254
7	freight_value	V_String	254

After

Record	Name	Type	Size
1	order_id	V_String	254
2	order_item_id	V_String	254
3	product_id	V_String	254
4	seller_id	V_String	254
5	shipping_limit_date	DateTime	19
6	price	Double	8
7	freight_value	Double	8

Figure 2: Utilising Auto Field for Brazilian E-Commerce Order Item Dataset

The Select tool keeps only the selected field (seller_id, seller_city, seller_state, seller_zip_code_prefix) in order to remove the unnecessary columns to keep the master table lean.

To enforce a strict data consistency and integrity across the item collections for the dataset (Orders streams, Order Items streams & Products streams), a Filter tool is utilised to discards the empty catalog lines (or NULL) where only rows with valid category classifications pass through the True Anchor. For example, *Figure 3* demonstrate the customised filter for Products streams to remove the 610 uncategorised products so category-level analysis in Power BI is accurate.

Figure 3: Customised filter expression for Products Dataset

The Unique tool is applied independently across the primary transaction keys (order_id) in the Order Items stream (olist_order_items_dataset.csv) and Orders stream (olist_orders_dataset.csv). Through the Unique anchor, it extracts the distinct lines by dropping any row duplication to protect metrics against double-counting before the next integration.

1.3 Feature Creation

After the data has been cleaned, Formula tools were used to create 2 new fields which are delivery_duration_days and order_month_year.

The formula used for delivery_duration_days is :

"delivery_duration_days = DateTimeDiff([order_delivered_customer_date], [order_purchase_timestamp], "days")"

This formula calculates the actual number of days from purchase to delivery, enabling logistics performance analysis. order_delivered_customer_date refers to the date the order was delivered to the customer and order_purchase_timestamp refers to the date and time the order was placed. DateTimeDiff is used to calculate the difference between the two dates.

As for order_month_year, the formula used is :

"order_month_year = DateTimeFormat([order_purchase_timestamp], "%Y-%m")"

This formula is used to convert the order purchase date and time into a specific text format (e.g., 2018-03). This is done to enable data visualisations to be done using the Line & Cluster Column Chart.

1.4 Data Integration

The workflow connects tables through a 3-stage cascading structure using sequential Join Tools to resolve relational schemas by combining the separate, isolated input vectors into a single, fully enriched master record set without relational data dropouts. In each Join tool's Select tab, the duplicate key column from the right input is deselected to prevent redundant columns in the output. For example the second order_id from Order Items.

- **First Join:** Blends the duplicate-free Orders stream (Left) and Order Items stream (Right) via the shared order_id field, connecting base orders directly with price logs.
- **Second Join:** Merges the output of the transaction join (Left) with the cleaned Products data stream (Right) using the product_id key, mapping category dimensions directly to sales lines.
- **Third Join:** Combining product-order stream (Left) with the Sellers lookup matrix (Right) using the seller_id field, appending vital regional attributes like seller_state and seller_city.

1.5 Data Aggregation

After integration, data aggregation was performed using the Summarise tool. The Summarize Tool is connected to the Join 3 output and aggregates the master table to produce a summary table with only key attributes needed for the data analysis and visualisation. The summary table consists of Seller State, Order Month Year, Product Category Name, Total Revenue, Total Freight Cost, Total Orders and Average Delivery Days fields. Table 2 shows the actions performed on the fields to produce the summary table.

Action	Field	Output Column Name
Group By	seller_state	Seller State
Group By	order_month_year	Order Month Year
Group By	product_category_name	Product Category Name
Sum	price	Total Revenue
Sum	freight_value	Total Freight Cost
Count	order_id	Total Orders
Average	delivery_duration_days	Average Delivery Days

Table 2 : Actions Performed on The Fields to Produce The Summary Table

1.6 Data Output

Once data aggregation was done, the data was exported into csv format using the Output Data tool. Two output files were created. One Output Data tool was connected directly from Join 3 to create an output dataset named "Olist Master Table.csv" which contains all of the data merged into Join 3. Another output dataset was exported named "Olist Aggregated Summary.csv" which was created by connecting the Output Data tool to the Summarise tool. This dataset contains the refined summary table, ideal for data visualisation.

2.0 Tools Used

This section briefly explains about all the tools used in the data workflow along with its purpose of use. Table 3 shows the summary of the tools used.

Tool	**Purpose**
Input Data	Load olist_orders, olist_order_items, olist_products, olist_sellers CSVs into the workflow
Auto Field	Convert the data types of fields across all datasets from string to appropriate data type
Filter	Exclude null values in olist_orders, olist_order_items and olist_products
Select	Rearrange the hierarchy of the dataset with seller_id, seller_city, seller_state, seller_zip_code_prefix fields in this stated order

Unique	Remove duplicate entries in olist_orders and olist_order_items dataset
Formula	Create two new fields named delivery_duration_days and order_month_year
Join	Merge all the datasets into one unified dataset
Summarise	Aggregate master table into summarised table with Seller State, Order Month Year, Product Category Name, Total Revenue, Total Freight Cost, Total Orders, Average Delivery Days fields
Output Data	Export two datasets named Olist Master Table and Olist Aggregated Summary into csv files

Table 3 : Summary of Tools Used Along The Data Workflow

3.0 Technical Challenges

- Duplicate Join keys across sequential Joins

  After Join 1, the output contains two copies of order_id from Orders and Order Items. If not addressed, Join 2 and Join 3 would carry forward the redundant columns, where it can cause ambiguity in the schema. This was resolved by using select configuration inside each Join tool to deselect the duplicate key from the right input before proceeding to the next stage.
- Choosing the right tools either Join or Union tools

  The first point of confusion was whether to use the Union tool or Join tool for data integration. The union tool combines tables vertically, while the Join tool combines table horizontally. Since each dataset contains different columns describing aspects of the same transaction, the Join tool was the best choice. Using Union would have produced a table with many null values.
- Date Format Parsing for order_month_year

  The order_purchase_timestamp field is stored as a full datetime string. Directly grouping by this field in the Summarize tool would produce one group per unique timestamp. The Formula tool was used to extract only the year-month portion using DateTimeFormat(), producing a clean string like ‘2018-03’ that allows useful monthly aggregation and supports Power BI’s visualization.